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Fundamentals of AI and ML Evaluated
Course Project

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INTRODUCTION :

In today's fast-paced and digitally connected world, mental health has emerged as a critical concern affecting individuals across all age groups. With the increasing use of digital platforms such as social media, messaging applications, and online forums, people often express their thoughts, emotions, and feelings through text. However, identifying emotional states from textual expressions manually can be challenging, subjective, and time-consuming. This creates a need for intelligent systems that can automatically analyze and interpret human emotions from written text.

The AI-Based Mental Health Text Analyzer is a machine learning-based project designed to address this need by analyzing user-input text and predicting the underlying emotional state. The system utilizes Natural Language Processing (NLP) techniques to process textual data and classify it into different emotional categories such as happiness, sadness, anger, and stress. This project not only demonstrates the practical application of artificial intelligence in understanding human emotions but also highlights the potential of technology in supporting mental well-being. By providing quick insights into emotional states, such systems can act as supportive tools in digital mental health monitoring and awareness.

Problem Statement :

In modern society, many individuals face challenges in expressing their emotions openly, and even when they do, it may not be easy for others to interpret their mental state accurately. Emotional distress, stress, and anxiety are often reflected subtly in the language people use, especially in text-based communication. However, there is currently a lack of simple, accessible tools that can automatically analyze such text and provide meaningful insights into a person's emotional condition.

The problem addressed in this project is the development of an automated system capable of detecting and classifying emotions from textual input using machine learning techniques.

The goal is to create a solution that can bridge the gap between human emotional expression and computational understanding, thereby enabling faster and more objective analysis of mental states.

This problem is particularly relevant in the context of increasing digital communication, where large volumes of text data are generated daily and manual analysis becomes impractical.

Objectives :

The primary objective of this project is to design and develop a system that can effectively analyze textual input and identify the underlying emotional state using machine learning techniques. The project aims to explore the application of Natural Language Processing in understanding human emotions and to implement a classification model that can categorize text into predefined emotional classes.

In addition to this, the project also seeks to build a simple and user-friendly interface that allows users to input their thoughts easily and receive immediate feedback in the form of predicted emotions. Another important objective is to evaluate the performance of the model and understand its limitations, particularly in handling complex or ambiguous text. Overall, the project is intended to provide a foundational understanding of how artificial intelligence can be applied to real-world problems related to mental health and emotional analysis.

Scope of the Project :

The scope of this project is limited to basic emotion detection from textual input using machine learning techniques. The system focuses on classifying text into a small set of predefined emotional categories, namely happiness, sadness, anger, and stress. It operates as a console-based application, making it simple and easy to implement while still demonstrating core concepts of NLP and machine learning. While the project successfully illustrates the process of emotion detection, it does not aim to provide a comprehensive psychological analysis or medical diagnosis.

The system is not designed to replace professional mental health services but rather to serve as an educational tool and a basic demonstration of AI capabilities. The scope can be expanded in the future to include more advanced features such as real-time analysis, integration with web applications, and the use of deep learning models for improved accuracy.

Literature Review :

The field of emotion detection from text has gained significant attention in recent years due to the growing importance of mental health awareness and the widespread use of digital communication. Early approaches to emotion detection relied heavily on rule-based systems, where specific keywords were associated with particular emotions. Although these methods were simple to implement, they lacked flexibility and often failed to capture the complexity of human language.

With advancements in artificial intelligence, machine learning techniques have become the preferred approach for emotion analysis. Algorithms such as Logistic Regression, Naive Bayes, and Support Vector Machines have been widely used for text classification tasks. Additionally, Natural Language Processing techniques like TF-IDF and word embeddings have enabled more accurate representation of textual data. More recently, deep learning models such as LSTM and transformer-based architectures like BERT have shown remarkable performance in understanding context and nuances in language.

This project adopts a machine learning-based approach using Logistic Regression and TF-IDF vectorization, providing a balance between simplicity and effectiveness while demonstrating key concepts in the field.

METHODOLOGY:

The methodology followed in this project involves a systematic process of data preparation, model training, and evaluation. Initially, a dataset of text samples labelled with corresponding emotions was created manually. This dataset serves as the foundation for training the machine learning model.

The next step involves preprocessing the text data to make it suitable for analysis. This includes converting all text to lowercase, removing unnecessary characters, and preparing the data for feature extraction. Once the text is cleaned, it is transformed into numerical form using TF-IDF vectorization, which captures the importance of words in the dataset.

After feature extraction, a Logistic Regression model is trained using the processed data. The model learns patterns associated with different emotions and uses this knowledge to classify new input text. Finally, the trained model is used to predict emotions for user-provided input, completing the workflow of the system.

System Architecture :

The system architecture of the Mental Health Text Analyzer is designed as a simple and efficient pipeline that processes user input and generates an output in the form of predicted emotion. The process begins with the user entering a text statement that reflects their thoughts or feelings. This input is then passed through a preprocessing stage where the text is cleaned and prepared for analysis.

Following preprocessing, the text is converted into numerical features using TF-IDF vectorization. These features are then fed into the trained machine learning model, which analyzes the patterns in the data and predicts the most likely emotional category. The final output is displayed to the user in a clear and understandable format. This structured flow ensures that the system operates efficiently while maintaining accuracy in predictions.

Tools and Technologies Used :

The development of this project relies on several tools and technologies that facilitate the implementation of machine learning and text analysis. Python is used as the primary programming language due to its simplicity and extensive support for data science libraries. Scikit-learn is utilized for implementing the machine learning model and handling tasks such as vectorization and classification.

The TF-IDF Vectorizer is used for converting textual data into numerical form, enabling the model to process and analyze it effectively. Logistic Regression is chosen as the classification algorithm due to its efficiency and suitability for text classification tasks. Additionally, TextBlob is used to provide basic sentiment analysis support, enhancing the overall functionality of the system. These technologies together form a robust framework for developing the Mental Health Text Analyzer.

Implementation :

The implementation of the Mental Health Text Analyzer is carried out in a step-by-step manner using Python. The process begins with importing the required libraries, followed by defining the training dataset consisting of labelled text samples. The dataset is then processed using TF-IDF vectorization to convert the text into numerical features.

Once the data is prepared, the Logistic Regression model is trained on the dataset. After training, the model is capable of predicting emotions for new input text. The user is prompted to enter a sentence, which is then processed and analyzed by the model. The predicted emotion is displayed as the output. The implementation is kept simple to ensure clarity while effectively demonstrating the core concepts of machine learning and NLP.

Testing and Results :

The AI-Based Mental Health Text Analyzer was tested using different types of input sentences to evaluate its accuracy and performance. The test cases included simple emotional statements, mixed-emotion sentences, and neutral text. Each input was processed through the system, and the predicted emotion was compared with the expected result.

The model performed well on simple and clearly expressed sentences such as “I am very happy today” or “I feel so sad and tired,” correctly identifying the corresponding emotions. This shows that the model has successfully learned basic patterns from the training data. However, it faced difficulties with complex or mixed-emotion sentences, where it often predicted only one dominant emotion.

Neutral sentences were also sometimes incorrectly classified since the model is designed to assign an emotion in every case.

Overall, the system achieved satisfactory results for a basic machine learning project. While it performs well on straightforward inputs, its accuracy can be improved by using a larger dataset and more advanced models. The testing phase helped identify limitations and provided direction for future improvements.

Challenges Faced :

During the development of this project, several challenges were encountered. One of the primary challenges was the limited size of the dataset, which affected the accuracy and generalization capability of the model. A small dataset makes it difficult for the model to learn diverse patterns, leading to incorrect predictions in some cases.

Another challenge was handling complex and ambiguous text, where emotions are not clearly defined or multiple emotions are present. The model often struggles to interpret such inputs accurately. Additionally, issues related to overfitting were observed, as the model tends to perform well on training data but may not generalize effectively to new inputs. These challenges highlight the need for larger datasets and more advanced models in future improvements.

Future Enhancements :

There are several ways in which this project can be enhanced to improve its performance and functionality. One of the most important improvements would be the use of a larger and more diverse dataset, which would enable the model to learn a wider range of linguistic patterns. Implementing advanced machine learning or deep learning models such as LSTM or BERT could significantly improve accuracy and the ability to understand context.

In addition to improving the model, the system can be extended by developing a web-based interface using frameworks like Streamlit or Flask, making it more interactive and user-friendly.

Features such as mood tracking, visualization of emotional trends, and personalized suggestions based on detected emotions can also be added. These enhancements would transform the project into a more comprehensive and practical application.

OUTPUT:

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS C:\Users\Anushka> cd OneDrive\Documents
PS C:\Users\Anushka\OneDrive\Documents> python emotion_detector.py

Enter your text (type 'exit' to stop): I am feeling angry right now

----- RESULT -----
Your Text: I am feeling angry right now
Detected Emotion: anger
Sentiment: Negative
Suggestion: Take deep breaths and calm down

Enter your text (type 'exit' to stop):
```

```
Enter your text (type 'exit' to stop): I am tired but still happy

----- RESULT -----
Your Text: I am tired but still happy
Detected Emotion: happy
Sentiment: Positive
Suggestion: Keep smiling and enjoy your day
```

```
Enter your text (type 'exit' to stop): I am stressed about exams

----- RESULT -----
Your Text: I am stressed about exams
Detected Emotion: stress
Sentiment: Neutral
Suggestion: Take a short break and relax

Enter your text (type 'exit' to stop):
```

Enter your text (type 'exit' to stop): Everything is going great

----- RESULT -----

Your Text: Everything is going great

Detected Emotion: happy

Sentiment: Positive

Suggestion: Keep smiling and enjoy your day

Enter your text (type 'exit' to stop): I am feeling lonely and upset

----- RESULT -----

Your Text: I am feeling lonely and upset

Detected Emotion: sad

Sentiment: Negative

Suggestion: Talk to someone you trust

Enter your text (type 'exit' to stop): I am so happy today

----- RESULT -----

Your Text: I am so happy today

Detected Emotion: happy

Sentiment: Positive

Suggestion: Keep smiling and enjoy your day

Enter your text (type 'exit' to stop):

Conclusion :

The AI-Based Mental Health Text Analyzer successfully demonstrates the application of machine learning and natural language processing in detecting emotions from text. The project provides valuable insights into how artificial intelligence can be used to interpret human language and emotions, contributing to the field of digital mental health.

Although the system has certain limitations due to its simplicity, it serves as a strong foundation for further development and exploration. The project emphasizes the importance of integrating technology with mental health awareness and showcases the potential of AI in creating supportive tools for emotional analysis and well-being.